

NORTHPORT PIER AT DEATHS DOOR, WI, USA

WMO: 994973

Lat: 45.2909N

Lon: 86.9768W

Elev: 617

StdP: 14.37

Time Zone: -6.00 (NAC)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-2.1	2.2	N/A	N/A	N/A	N/A	N/A	N/A	34.0	28.6	30.5	22.3	9.2	290	0.557

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.7	81.6	N/A	78.3	N/A	75.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	6.0	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 30.0	26.2	23.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	45.2	21.3	20.4	30.6	40.3	50.7	61.1	69.1	68.2	61.8	50.4	38.3	28.6		
	DBStd	18.44	10.36	10.27	9.28	5.81	6.67	5.73	5.41	3.97	6.04	6.64	7.51	8.52		
	HDD50	3784	891	828	603	292	74	1	0	0	1	78	354	662		
	HDD65	7549	1356	1248	1067	740	446	145	20	14	131	455	800	1127		
	CDD50	2037	0	0	1	2	94	334	593	564	355	90	4	0		
	CDD65	327	0	0	0	0	1	27	149	113	35	2	0	0		
	CDH74	956	0	0	0	0	4	75	582	228	65	2	0	0		
	CDH80	161	0	0	0	0	0	9	118	25	9	0	0	0		
(14) Wind	WSAvg	9.0	10.1	9.2	8.5	9.7	8.6	7.0	6.9	6.8	8.5	10.6	10.8	11.0		
(15) Precipitation	PrecAvg	28.7	1.6	1.3	1.9	2.4	2.8	3.3	2.6	2.5	3.3	3.0	2.4	1.5		
	PrecMax	37.1	4.4	2.5	5.0	4.3	4.6	8.7	7.0	4.0	7.8	7.1	4.8	3.0		
	PrecMin	22.9	0.4	0.4	0.3	0.5	1.1	0.5	0.7	0.6	1.5	0.8	0.6	0.2		
	PrecStd	3.4	1.1	0.5	1.3	1.1	1.1	2.0	1.5	1.0	1.5	1.5	1.1	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.1	42.3	58.9	59.3	73.2	80.7	88.1	83.4	80.0	71.0	56.6	49.0		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	41.2	38.2	51.3	54.7	68.0	75.8	83.7	79.5	75.6	65.3	53.4	43.9		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	38.1	35.7	46.0	51.7	64.7	72.5	80.1	76.5	72.4	62.3	50.7	41.7		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	35.2	33.5	42.2	48.5	61.3	69.9	77.0	74.3	69.9	59.7	48.7	39.8		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.4	11.3	12.0	10.5	12.5	11.9	11.7	10.2	10.5	8.6	8.1	8.3		
		MCDBR	11.7	12.4	15.5	15.8	18.5	15.9	15.3	13.4	13.5	12.3	10.1	9.7		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.270	0.286	0.319	0.356	0.387	0.393	0.385	0.388	0.366	0.335	0.304	0.275		
	taud		2.513	2.432	2.405	2.378	2.360	2.390	2.429	2.414	2.467	2.527	2.539	2.534		
	Ebn at Noon		274	287	290	287	280	278	279	273	271	266	257	259		
	Edh at Noon		21	27	32	36	38	37	35	35	30	25	20	18		
(44) All-Sky Solar Radiation	RadAvg		422	732	1133	1465	1788	1926	1980	1676	1284	779	446	326		
	RadStd		41	83	106	135	144	150	81	111	97	82	46	32		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (47 neighbors)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page